In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
sns.set(style="whitegrid")
plt.style.use('seaborn')
```

In [3]:
```
titanic_df = sns.load_dataset('titanic')
```

In [4]:
```
print(titanic_df.head())
```

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

In [5]:
```
print(titanic_df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

In [6]:
```
print(titanic_df.isnull().sum())
```

```
survived         0
pclass           0
sex              0
```

```
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

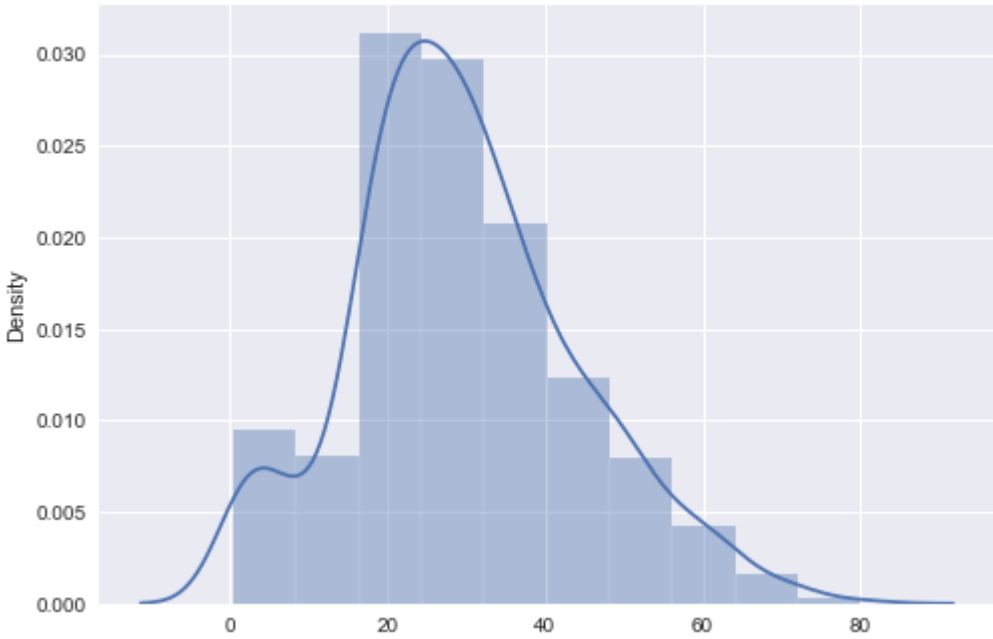
In [8]:

```
sns.distplot(x = titanic_df['age'], bins = 10)
```

```
C:\Users\kolig\anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWarni
ng: `distplot` is a deprecated function and will be removed in a future version. Plea
se adapt your code to use either `displot` (a figure-level function with similar flex
ibility) or `histplot` (an axes-level function for histograms).
  warnings.warn(msg, FutureWarning)
```

Out[8]:

```
<AxesSubplot:ylabel='Density'>
```

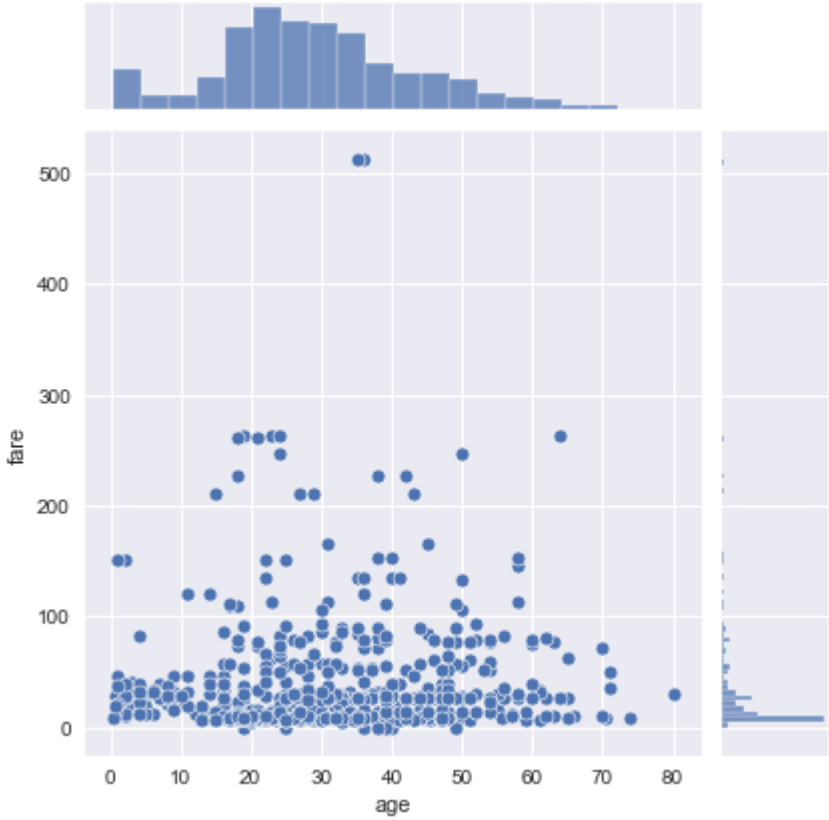


In [9]:

```
sns.jointplot(x = titanic_df['age'], y = titanic_df['fare'], kind ='scatter')
```

Out[9]:

```
<seaborn.axisgrid.JointGrid at 0x1bbc15edfa0>
```

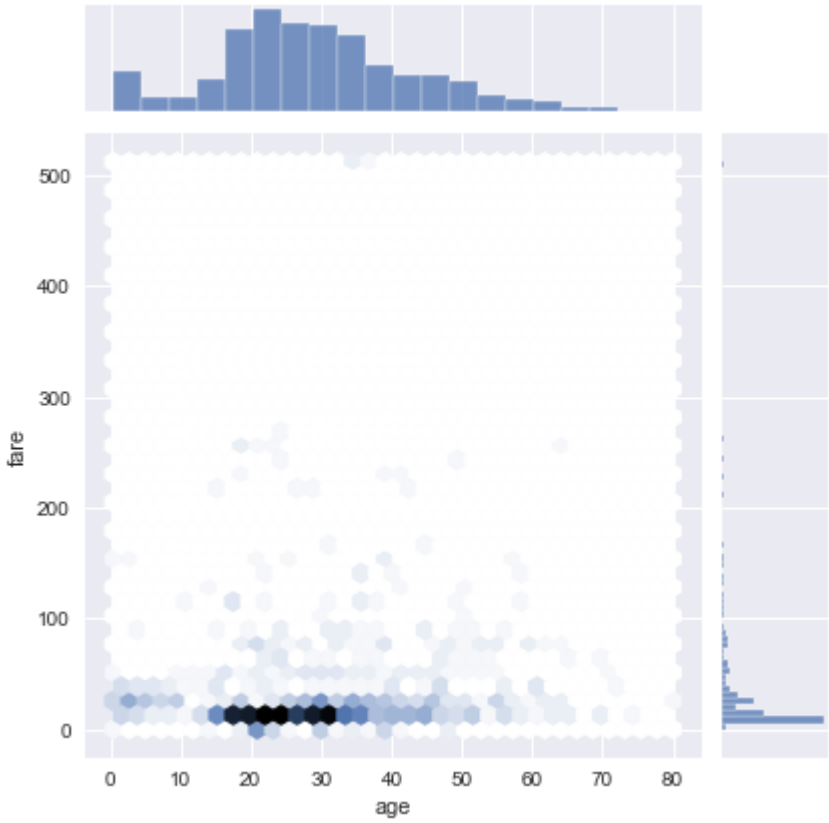


In [10]:

```
sns.jointplot(x = titanic_df['age'], y = titanic_df['fare'], kind ='hex')
```

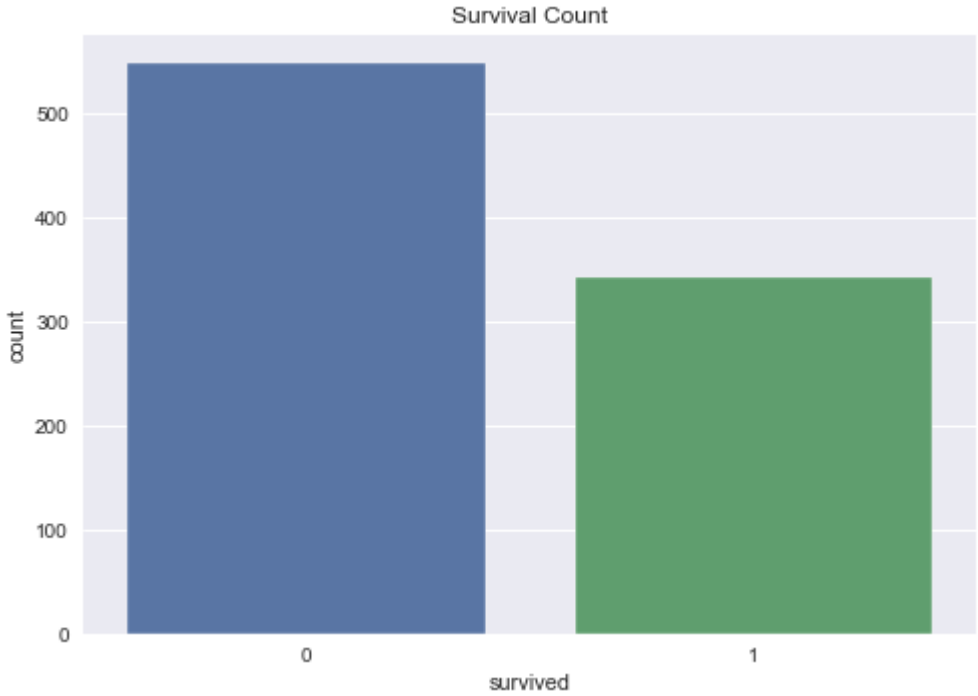
Out[10]:

```
<seaborn.axisgrid.JointGrid at 0x1bbc1c0dd60>
```



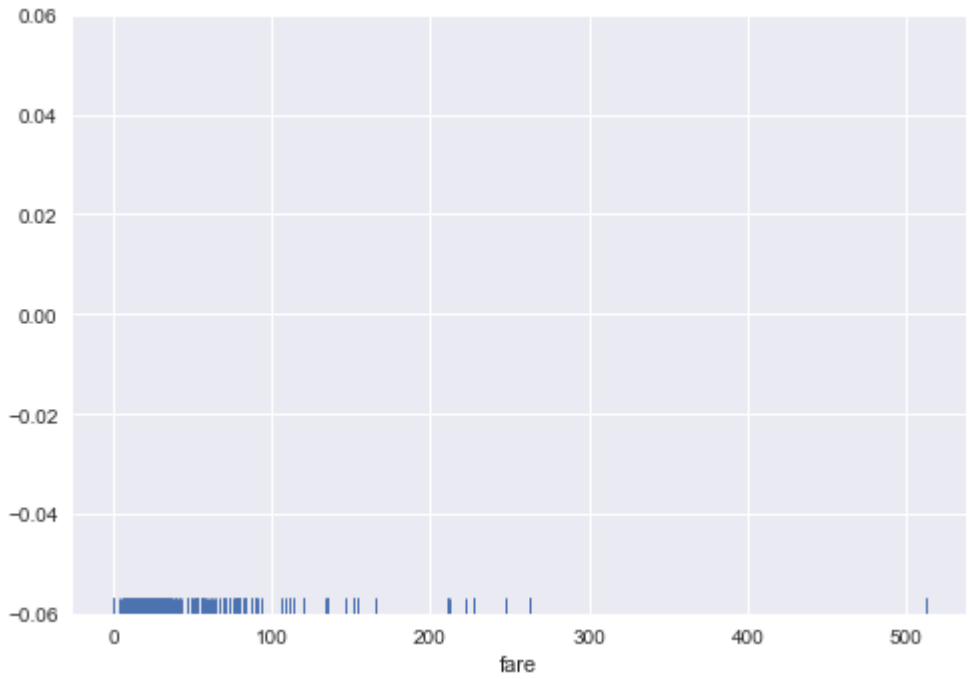
In [11]:

```
sns.countplot(x='survived', data=titanic_df)
plt.title("Survival Count")
plt.show()
```



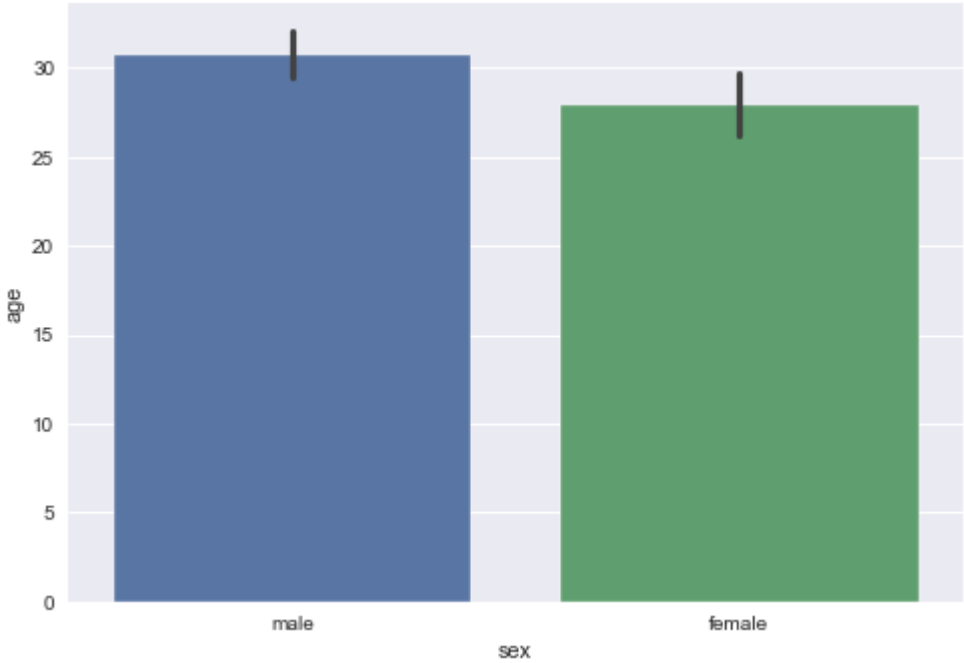
In [13]:
```
sns.rugplot(titanic_df['fare'])
```

Out[13]:
```
<AxesSubplot:xlabel='fare'>
```



In [14]:
```
sns.barplot(x='sex', y='age', data=titanic_df)
```

Out[14]:
```
<AxesSubplot:xlabel='sex', ylabel='age'>
```

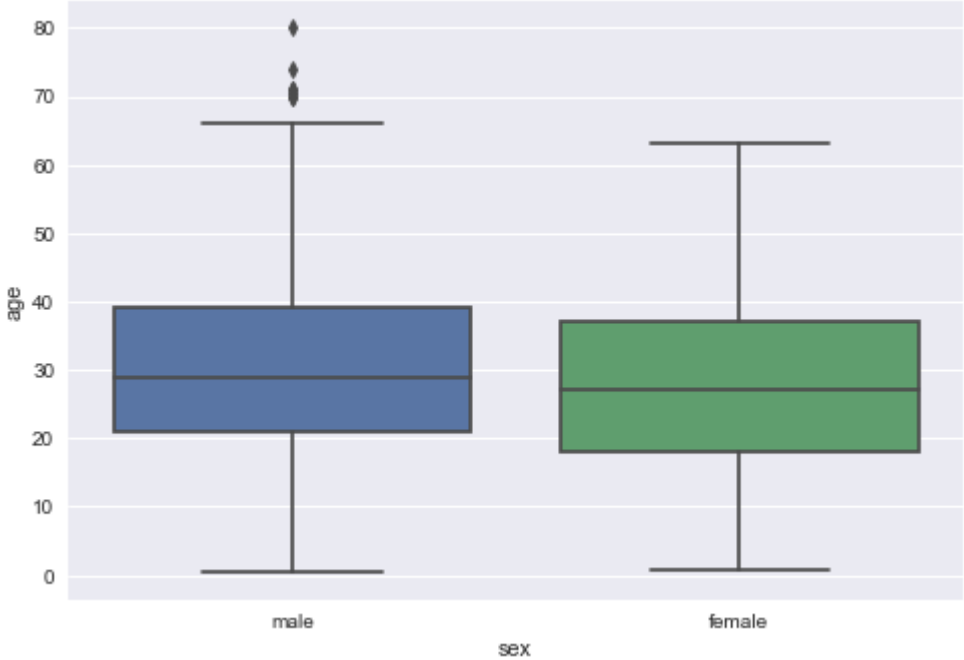


In [15]:

```
sns.boxplot(x='sex', y='age', data=titanic_df)
```

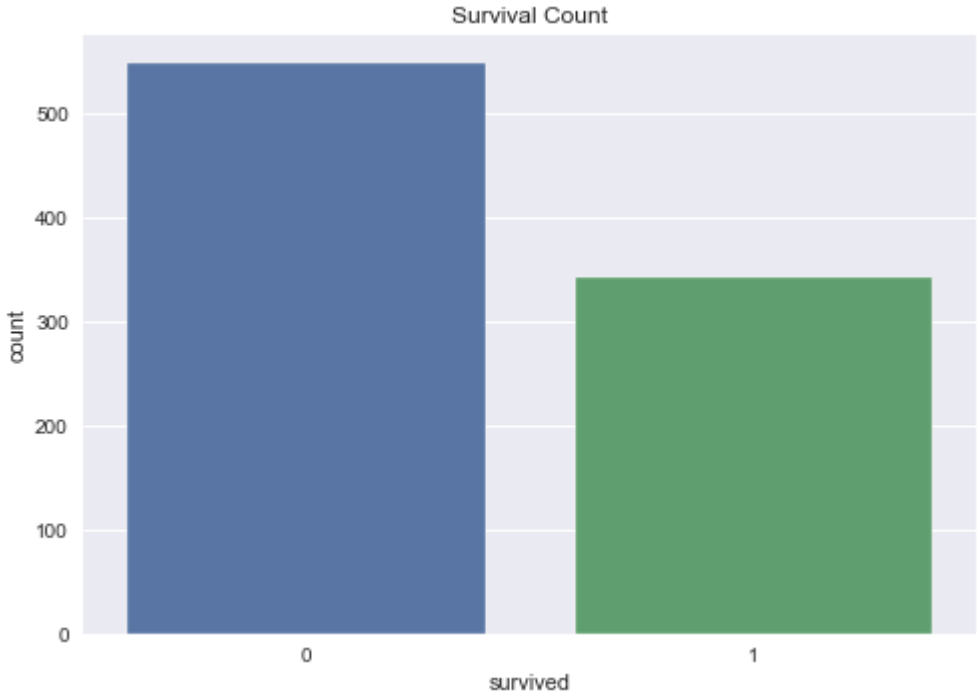
Out[15]:

```
<AxesSubplot:xlabel='sex', ylabel='age'>
```



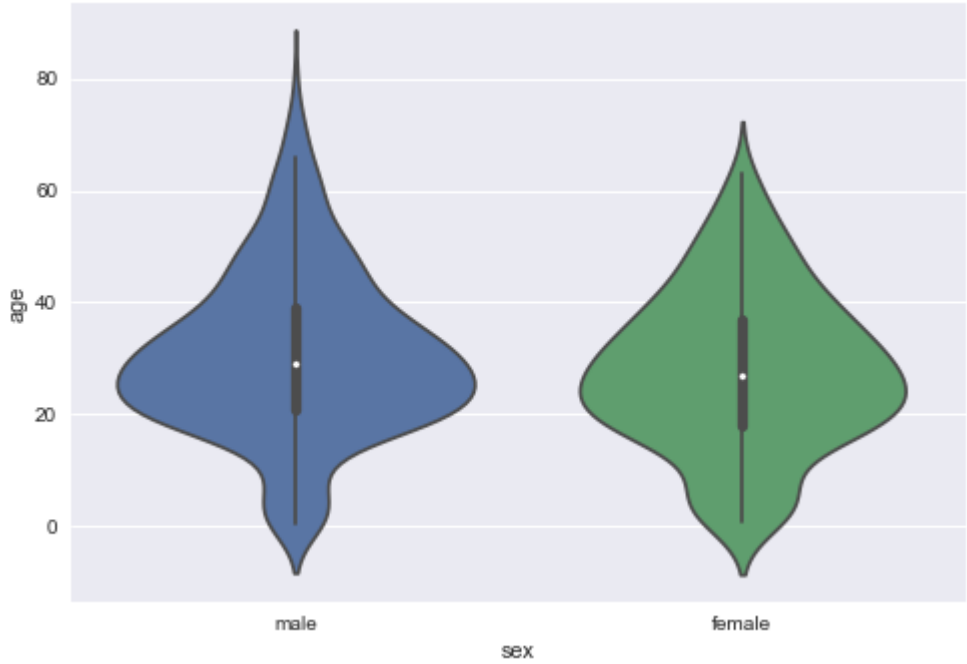
In [16]:

```
sns.countplot(x='survived', data=titanic_df)
plt.title("Survival Count")
plt.show()
```



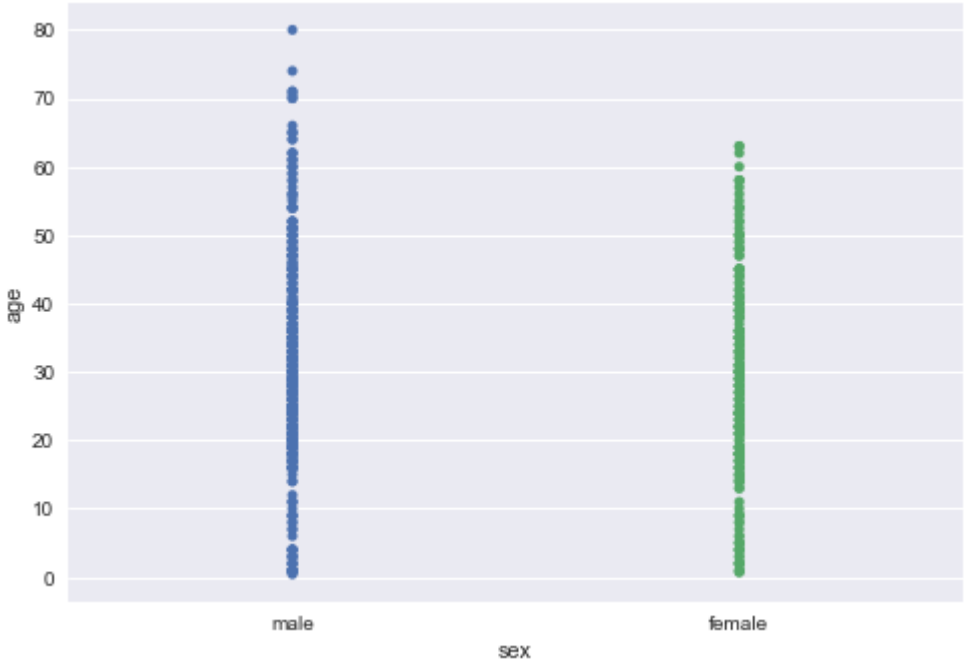
In [17]:
```
sns.violinplot(x='sex', y='age', data=titanic_df)
```

Out[17]:
```
<AxesSubplot:xlabel='sex', ylabel='age'>
```



In [18]:
```
sns.stripplot(x='sex', y='age', data=titanic_df, jitter=False)
```

Out[18]:
```
<AxesSubplot:xlabel='sex', ylabel='age'>
```

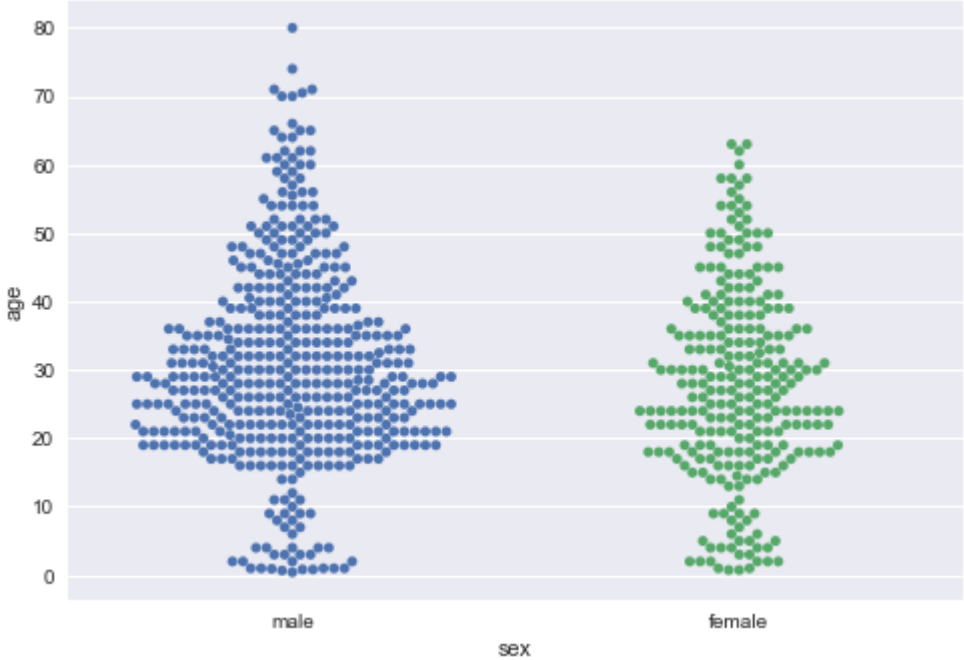


In [19]:

```
sns.swarmplot(x='sex', y='age', data=titanic_df)
```

Out[19]:

```
<AxesSubplot:xlabel='sex', ylabel='age'>
```



In [20]:

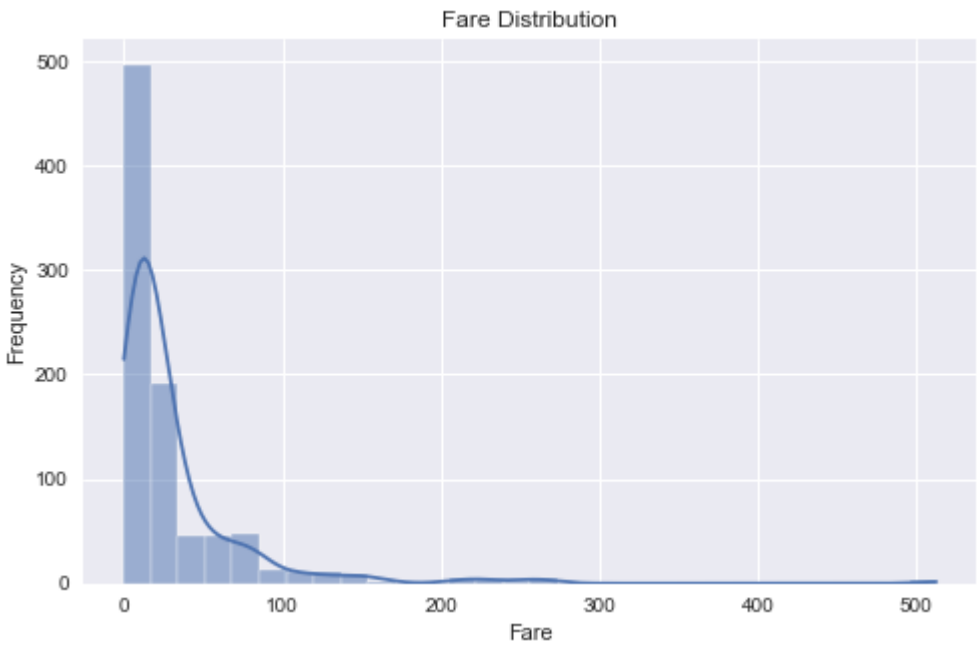
```
corr = titanic_df.corr()
sns.heatmap(corr, annot=True)
```

Out[20]:

```
<AxesSubplot:>
```

In [22]:

```
plt.figure(figsize=(8,5))
sns.histplot(titanic_df['fare'], bins=30, kde=True)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.show()
```



In [ ]: